A Minor Project Synopsis

On

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Proactive Emotion Tracker

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1. Abstract

This project presents a **Proactive Emotion Tracker** — an AI-powered, real-time affective computing system designed to continuously monitor, classify, and respond to a user's emotional state using multimodal signals: facial expression analysis, voice tone detection, and text sentiment analysis. The system employs a hybrid deep-learning architecture combining Convolutional Neural Networks (CNN) for facial affect recognition, a fine-tuned BERT model for text-based sentiment extraction, and acoustic feature analysis (MFCC/pitch) for speech emotion recognition. Emotion predictions are fused through a weighted ensemble meta-classifier, producing a unified affective state label across seven core emotions: *Neutral, Happy, Sad, Angry, Fearful, Disgusted,* and *Surprised*. A FastAPI backend delivers real-time predictions and personalized mental wellness recommendations through a React-based dashboard. The system proactively alerts users and counsellors when prolonged negative emotional states are detected, bridging AI research with practical mental health support.

Keywords — Emotion Recognition, Affective Computing, CNN, BERT, MFCC, Sentiment Analysis, FastAPI, Mental Health, Multimodal AI, Real-Time Processing.

2. Introduction

Emotional well-being is a cornerstone of human productivity, academic performance, and overall health. Despite growing awareness of mental health, most individuals lack access to continuous, objective monitoring of their emotional states. Current solutions are either reactive (therapy sessions, mood-logging apps) or limited to single-modality sensing.

Advances in deep learning and multimodal sensing have made it possible to infer human emotions non-intrusively from facial geometry, vocal patterns, and textual expression — the three most natural channels of emotional communication. However, integrating these modalities into a single, proactive, real-time system that generates actionable wellness interventions remains an open engineering challenge.

This project implements exactly that: a **Proactive Emotion Tracker** that monitors users through their camera, microphone, and text input; classifies their emotional state in real time; detects prolonged negative affect patterns; and delivers personalized recommendations. The system is deployed as a live FastAPI service with a React frontend, making it an end-to-end deployable mental wellness tool — not merely an academic prototype.

3. Problem Statement

The following specific challenges motivated this implementation:

- Single-modality emotion recognition systems suffer from poor accuracy in real-world conditions (lighting changes, noise, ambiguous text) and cannot cross-validate emotional signals.
- Existing mobile mental health apps rely entirely on self-reporting, introducing subjective bias and retrospective recall errors.
- Students and professionals experiencing emotional distress often go undetected in digital learning or remote-work environments where physical cues are absent.
- No lightweight, deployable API currently combines facial, vocal, and textual emotion streams into a unified, proactive wellness alert system.
- The absence of explainable emotion feedback means users cannot understand which signal drove a given assessment, reducing trust and actionability.

4. Objectives

- Build a **CNN-based facial emotion recognition** model achieving $\geq 85\%$ accuracy on the FER-2013 benchmark across seven emotion classes.
- Develop a **BERT-based text sentiment classifier** fine-tuned on GoEmotions achieving F1-macro ≥ 0.80.
- Implement an **acoustic emotion classifier** using MFCC, chroma, and spectral features achieving $\geq 78\%$ accuracy on RAVDESS.
- Design a **multimodal fusion layer** that combines the three modality outputs using weighted voting to produce a single, reliable emotion prediction.
- Build and deploy a **FastAPI (Uvicorn)** REST API that accepts video frames, audio clips, and text; returns unified emotion label, confidence, and wellness recommendations.
- Implement **proactive alerting logic** that detects prolonged negative emotion sequences (> 5 min) and triggers counsellor notifications.

5. Literature Review

5.1 Traditional Emotion Recognition Models

Ekman and Friesen (1978) established the universality of six basic facial expressions, providing the psychological foundation for automatic affect recognition [1]. Logistic-regression and SVM-based classifiers operated on hand-crafted features (HOG, LBP) and dominated early work. However, El Ayadi et al. (2011) showed these approaches plateau rapidly on real-world, in-the-wild data [4].

5.2 Deep Learning for Facial and Speech Emotion

Goodfellow et al. (2013) introduced the FER-2013 dataset and demonstrated CNN-based facial recognition [2]. Li & Deng (2022) surveyed deep FER, identifying occlusion, illumination, and subject variability as primary challenges [3]. For speech, Livingstone & Russo (2018) released RAVDESS — 1,440 recordings across 8 emotions from 24 professional actors — which serves as the primary audio benchmark for this work [5].

5.3 BERT and Text Sentiment

Devlin et al. (2019) introduced BERT, achieving state-of-the-art performance across NLP tasks including sentiment classification [6]. Demszky et al. (2020) released GoEmotions — a 58 k Reddit comment dataset annotated with 27 emotion categories — enabling fine-grained text emotion detection [7].

5.4 Multimodal Emotion Fusion

Shenoy & Sardana (2020) demonstrated that late fusion of facial, speech, and text modalities outperforms any single-modality approach by 6–12 percentage points on IEMOCAP [8]. Hazarika et al. (2018) proposed conversational memory networks for context-aware multimodal fusion [9].

5.5 Research Gap Addressed

Reference	Method	Modalities	Limitation
Goodfellow et al. (2013)	CNN on FER-2013	Face only	No multimodal; no deployment
El Ayadi et al. (2011)	MFCC + SVM	Voice only	Single modality; offline only
Devlin et al. (2019)	BERT	Text only	No real-time API; no fusion
Shenoy & Sardana (2020)	Late fusion	Face+Voice+Text	No live deployment; no alerts
This Work (2026)	CNN+BERT+MFCC+API	All three	Full-stack; proactive alerts

Table 1: Comparison of related work and the research gap addressed by this project.

6. Methodology

The implementation follows a five-phase pipeline, each phase feeding into the next.

6.1 Phase 1 — Data Preparation

Three benchmark datasets are used: **FER-2013** (35,887 grayscale facial images, 7 classes), **RAVDESS** (1,440 audio recordings, 8 emotions), and **GoEmotions** (58,009 Reddit comments, 27 emotions collapsed to 7 core classes). Data augmentation (flipping, rotation, noise injection) is applied to each modality to address class imbalance.

6.2 Phase 2 — Model Training

Facial CNN: A ResNet-50 backbone pre-trained on ImageNet is fine-tuned on FER-2013. The classification head is replaced with a 7-class softmax layer trained with Adam (lr = 1e-4) and early stopping.

Text BERT: `bert-base-uncased` is fine-tuned on GoEmotions with lr = 2e-5, batch

size 32, and 4 epochs. Token sequences are padded/truncated to 128 tokens.

Speech LSTM: 40 MFCC coefficients, chroma features, and spectral contrast are extracted via `librosa`. A 3-layer bidirectional LSTM classifies feature vectors into 7 emotion categories.

6.3 Phase 3 — Multimodal Fusion

A weighted late-fusion strategy combines the three modality probability vectors:

$$P_{\text{fused}} \;=\; w_f \cdot P_{\text{face}} \;+\; w_t \cdot P_{\text{text}} \;+\; w_s \cdot P_{\text{speech}}$$

where weights w_f=0.45, w_t=0.35, w_s=0.20 are optimized on the IEMOCAP validation set. The final emotion label is $\arg\max(P_{\text{fused}})$.

6.4 Phase 4 — Proactive Alerting

A sliding window (5-minute duration, 1-minute step) tracks the dominant emotion sequence. If negative-valence emotions (Sad, Angry, Fearful) persist for ≥ 5 consecutive minutes, an alert is dispatched via WebSocket to the counsellor dashboard and an in-app recommendation card is generated.

6.5 Phase 5 — API Deployment

All models and the fusion layer are serialized with `torch.save` and `joblib`. FastAPI exposes `POST /predict/emotion`, `GET /history/{user_id}`, and `WebSocket /alerts`. The backend runs on Uvicorn with hot-reload in development and multi-worker mode in production.

7. Tools & Technologies

Category	Technology	Purpose
Language	Python 3.10+	Core ML pipeline and backend API
Deep Learning	PyTorch 2.0	CNN, LSTM, and BERT fine-tuning
NLP	HuggingFace Transformers	BERT tokenization and fine-tuning
Audio	Librosa 0.10	MFCC, chroma, spectral features
Computer Vision	OpenCV + dlib	Real-time face detection and alignment
API Framework	FastAPI + Uvicorn	REST and WebSocket serving
Frontend	React + Chart.js	Live dashboard and emotion timeline
Database	SQLite / PostgreSQL	Emotion history and user profiles
Serialization	Torch.save / Joblib	Model and pipeline persistence
Version Control	Git / GitHub	Code collaboration and management
Dev Environment	VS Code / Jupyter	Development and experimentation

Table 2: Tools and technologies used in the Proactive Emotion Tracker.

8. Implementation & System Architecture

8.1 System Architecture Overview

The system is structured as three cooperative layers:

1. **Sensing Layer:** The browser or mobile client captures webcam frames at 5 fps, microphone audio in 2-second windows, and user-typed text. Raw data is streamed to the FastAPI backend via HTTP multipart or WebSocket.
2. **Intelligence Layer:** The three independent classifiers (Facial CNN, Text BERT, Speech LSTM) run in parallel, each producing a probability vector over 7 emotion classes. The fusion module combines outputs and applies the proactive alerting logic.
3. **Presentation Layer:** The React dashboard visualises a real-time emotion gauge, a 24-hour emotion timeline chart, modality attribution bars, and wellness recommendation cards. A separate counsellor portal receives WebSocket alerts with timestamps and session context.

8.2 API Endpoint Summary

Method	Endpoint	Description
POST	/predict/emotion	Accepts frame, audio and text; returns emotion label, confidence and recommendations
GET	/history/{id}	Returns paginated emotion history with timestamps
POST	/alert/configure	Set alert thresholds per user (duration, emotion classes)
WS	/alerts	WebSocket stream for real-time counsellor notifications
GET	/health	Service liveness probe for monitoring

Table 3: FastAPI endpoint summary for the Proactive Emotion Tracker.

9. Expected Results & Evaluation Metrics

Based on literature benchmarks and pilot experiments, the following target performance metrics are expected:

Metric	Target	Benchmark	Description
Facial CNN Accuracy	$\geq 85\%$	FER-2013	7-class emotion classification
Text BERT F1-Macro	≥ 0.80	GoEmotions	Across all 7 emotion classes
Speech LSTM Accuracy	$\geq 78\%$	RAVDESS	7-class audio emotion
Fused Model Accuracy	$\geq 88\%$	IEMOCAP subset	Multimodal late fusion result
API Latency (P95)	≤ 200 ms	—	End-to-end prediction time
Alert Precision	$\geq 90\%$	—	True distress alert rate

Table 4: Target evaluation metrics for each system component.

Each model is evaluated on its respective held-out test split. The fusion system is additionally evaluated on a 500-sample multimodal test set collected via controlled user sessions. Confusion matrices, Precision-Recall curves, and modality-contribution attribution plots are generated for the final report.

10. Applications

- **University Mental Health Monitoring:** Passive monitoring of student emotional states during online lectures and exams; automatic referral to counsellors when prolonged distress is detected.

- **Corporate Employee Wellness:** Integration with video-conferencing tools to flag emotional burnout in remote teams; HR dashboard with anonymised aggregate wellness reports.
- **E-Learning Platforms:** Adaptive content delivery based on detected learner frustration or disengagement; real-time difficulty adjustment of course material.
- **Healthcare & Therapy:** Objective emotion timeline data supplementing therapist sessions; monitoring patients with depression or anxiety disorders between clinical visits.
- **Customer Experience Analytics:** Sentiment-aware virtual agents that adapt conversational tone to the user's detected emotional state in real time.
- **Accessibility Tools:** Assistive applications for users with alexithymia providing real-time self-awareness feedback on their own emotional expressions.

11. Conclusion & Future Scope

11.1 Conclusion

This project delivers a robust, full-stack **Proactive Emotion Tracker** that addresses a genuine societal need — the objective, continuous monitoring of emotional well-being in digital environments. By combining three complementary modalities (facial expression, vocal tone, textual sentiment) in a weighted late-fusion framework, the system is designed to substantially outperform any single-modality approach. The live FastAPI deployment, React dashboard, and WebSocket-based counsellor alerting transform this from an academic experiment into a deployable mental wellness tool suitable for university campuses, corporate environments, and healthcare settings.

The project contributes a novel proactive alerting mechanism that moves emotion recognition from a passive reporting tool to an active intervention system — notifying support personnel before a distress event escalates.

11.2 Future Scope

- **Physiological Integration:** Incorporate heart rate variability (wearables) and galvanic skin response as additional input modalities for more precise arousal estimation.
- **Transformer-Based Audio:** Replace MFCC-LSTM with a `wav2vec 2.0`-based speech emotion encoder for improved contextual audio understanding.
- **Privacy-Preserving Learning:** Apply federated learning to train models on user devices without transmitting raw biometric data to central servers.
- **Cultural Calibration:** Build culture-specific datasets to address cross-cultural variability in facial and vocal emotion display rules across different demographics.
- **Edge Deployment:** Quantize and prune models for on-device deployment on mobile and embedded hardware (Raspberry Pi, smartphones) enabling fully offline operation.
- **Longitudinal Wellness Analytics:** Add long-term trend analysis (weekly and

monthly patterns) with personalised intervention recommendations based on individual user history.

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